

Representación de posiciones y momentums

Estado estacionario:

$$\psi(x, t) = \psi_p(x) \phi(t) \sim e^{\frac{i}{\hbar} p x} \underbrace{e^{-\frac{i}{\hbar} E t}}$$

$$|\psi|^2 = 1$$

$$\int_{-\infty}^{\infty} |\psi|^2 dx \sim \infty \left. \begin{array}{l} \text{No es aplicable} \\ \text{T. Ehrenfest} \end{array} \right\}$$

¿Qué significa esto?

Una partícula real no puede tener momentum definido.

¿Cuál es la distribución de momentums?

$$\psi_p = A e^{\frac{i}{\hbar} p x}$$

$$(\psi_{p'}, \psi_p) = \int_{-\infty}^{\infty} dx A^* e^{-\frac{i}{\hbar} p' x} A e^{\frac{i}{\hbar} p x} = |A|^2 \int_{-\infty}^{\infty} dx e^{\frac{i}{\hbar} (p-p') x}$$

Se conoce:

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} dx \exp(ikx) = \delta(k) \quad \text{(Delta de Dirac)}$$

$$(\psi_{p'}, \psi_p) = 2\pi |A|^2 \delta\left(\frac{p-p'}{\hbar}\right)$$

$$\left. \begin{array}{l} (\psi_{p'}, \psi_p) = 2\pi |A|^2 \hbar \delta(p-p') \\ |A| = \frac{1}{\sqrt{2\pi \hbar}} \end{array} \right\}$$

Onda plana normalizada:

$$\psi_p = \frac{1}{\sqrt{2\pi \hbar}} e^{\frac{i}{\hbar} p x}$$

$$\psi(x, t) = \int_{-\infty}^{\infty} dp \underbrace{\phi(p, t)}_{\text{Coeficientes de Fourier}} \psi_p(x)$$

$$\begin{aligned}
 (\psi_p(x), \psi_{(x,t)}) &= \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dp' \phi \psi_{p'(x)}^* \psi_{p'}(x) \\
 &= \frac{1}{2\pi\hbar} \int dx \int dp \phi \underbrace{e^{\frac{i}{\hbar}(p'-p)x}} \\
 &= \int dp \phi(p', t) \delta(p' - p) \\
 &= \phi(p, t)
 \end{aligned}$$

$$\rightarrow \phi(p, t) = \int dx \psi_p^*(x) \psi_{(x,t)}$$

$$\phi \equiv \phi(p', t)$$

$$\begin{aligned}
 \psi_{(x,t)} &= \int_{-\infty}^{\infty} dp \phi(p, t) \psi_p(x) \quad \text{Coeficientes de Fourier} \\
 \psi_p &= \frac{1}{\sqrt{2\pi\hbar}} e^{\frac{i}{\hbar}px}
 \end{aligned}$$

$$\phi(p, t)$$

$$\phi(p, t) = \int dx \psi_p^*(x) \psi(x, t)$$

$$\phi(p, t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx e^{-\frac{i}{\hbar}px} \psi(x, t)$$

$$\begin{aligned}
 \bullet \int_{-\infty}^{\infty} dp \phi^*(p, t) \phi(p, t) &= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dp \int dx e^{\frac{i}{\hbar}px} \psi^*(x, t) \int dy e^{-\frac{i}{\hbar}py} \psi(y, t) \\
 &= \frac{1}{2\pi\hbar} \int dp dx dy \psi^*(x, t) \psi(y, t) e^{\frac{i}{\hbar}(px - py)}
 \end{aligned}$$

$$\int_{-\infty}^{\infty} dp \phi^*(p, t) \phi(p, t) = \int dx dy \psi^*(x, t) \cdot \psi(y, t) \delta(x - y) = \int dx |\psi|^2 = 1$$

$|\phi|^2$: Densidad de probabilidad del momentum

$|\phi|^2 dp$: Probabilidad de que el momentum se encuentre entre p y $p + dp$.

$$\psi(x, t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp e^{\frac{i}{\hbar}px} \phi(p, t)$$

$$\downarrow$$

$$|\psi|^2$$

$$\psi(x, t)$$

$$\downarrow$$

$$|\phi|^2$$

$$\phi(p, t)$$

(el valor medio depende del estado)

$$\langle P \rangle_{\psi} \equiv \frac{\hbar}{i} \int dx \psi^*(x, t) \frac{\partial}{\partial x} \psi(x, t)$$

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$$\langle P \rangle_\psi = \frac{\hbar}{i} \int dx \psi^*(x,t) \frac{\partial}{\partial x} \psi(x,t)$$

$$\psi(x,t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp e^{\frac{i}{\hbar} p x} \phi(p,t)$$

$$= \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dp \int_{-\infty}^{\infty} dp' \phi^*(p,t) \dots$$

$$e^{-\frac{i}{\hbar} p x} p' \phi(p',t) e^{\frac{i}{\hbar} p' x}$$

$$= \int_{-\infty}^{\infty} dp \int_{-\infty}^{\infty} dp' \phi^*(p,t) \phi(p',t) p' \delta(p'-p)$$

$$= \int_{-\infty}^{\infty} dp \phi^*(p,t) \phi(p,t) p = \int_{-\infty}^{\infty} dp p |\phi|^2$$

$$\langle P \rangle_\psi = \int_{-\infty}^{\infty} dp p |\phi|^2$$

$$\int_{-\infty}^{\infty} dp \underbrace{\phi^* p \phi}_{\phi^* P \phi}$$

$\langle P \rangle$ en el espacio momentum actúa como multiplicativo.

$$\langle R \rangle_\psi = \int dx \psi^* x \psi$$

$$= \frac{1}{2\pi\hbar} \int dx \int dp e^{-\frac{i}{\hbar} p x} \phi^*(p,t) x \dots$$

$$\int dp' e^{\frac{i}{\hbar} p' x} \phi(p',t)$$

$$= \frac{1}{2\pi\hbar} \int dp dp' \phi_p^* \phi_{p'} \int dx e^{-\frac{i}{\hbar} p x} \underbrace{x e^{\frac{i}{\hbar} p' x}}_{\text{circulo}}$$

$$x e^{\frac{i}{\hbar} p' x} = \left(\frac{\partial}{\partial p'} e^{\frac{i}{\hbar} p' x} \right) \frac{\hbar}{i}$$

$$\frac{\hbar}{i} \frac{\partial}{\partial p'} \text{ (operador de posición en el espacio momentum)}$$

(sin embargo es $-\frac{\hbar}{i} \frac{\partial}{\partial p'}$)

$$* H_\psi = (\psi, H\psi)$$

(¿Cómo opera H si ψ está en el espacio momentum?)

(Sol. De la ec. de Sch estacionaria)

$$\psi_{(x,t)} = \sum_n \alpha_n \underbrace{\psi_n(x)}_{\text{estado estac.}} e^{-\frac{i}{\hbar} E_n t}$$

$$\langle H \rangle_\psi = (\psi, H\psi)$$

$$\langle H \rangle_\psi = i \hbar \int_{-\infty}^{\infty} dx \psi^* \frac{\partial}{\partial t} \psi$$

$$i \hbar \frac{\partial \psi}{\partial t} = H\psi$$

(No significa que $H = i \hbar \frac{\partial}{\partial t}$ (ψ se determina))

$$\langle H \rangle_\psi = \int_{-\infty}^{\infty} dx \sum_{n,m} \alpha_n^* \alpha_m \psi_n^*(x) \psi_m(x) e^{\frac{i}{\hbar} E_n t} E_m e^{-\frac{i}{\hbar} E_m t}$$

Como $\int_{-\infty}^{\infty} dx \psi_n^*(x) \psi_m(x) = \delta_{nm}$

$$\Rightarrow \langle H \rangle_\psi = \sum_n |\alpha_n|^2 E_n$$

$|\alpha_n|^2$: probabilidad de que la energía tenga el valor E_n .

ψ contiene toda la información

ψ, ϕ, α_n información en diferentes bases.

Ejemplo

$$\psi_{(x,t)} = A e^{-\frac{|x|}{L} + \frac{i P_0 x}{\hbar}}$$

1) Normalizar $A = 1/\sqrt{L}$

2) R_ψ $R_\psi = 0$

3) P_ψ $P_\psi = P_0$

4) $\phi(P)$

5) R_ϕ

6) P_ϕ

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx \psi(x) e^{-\frac{i}{\hbar} p x}$$

$$\phi(p) = \frac{2L\hbar^2}{\sqrt{2\pi\hbar L}} - \frac{1}{L^2(p_0 - p)^2 + \hbar^2}$$

Verificar: $\int_{-\infty}^{\infty} |\phi|^2 dp = 1$

- $R_\phi = \frac{\hbar}{i} \int_{-\infty}^{+\infty} \phi^* p \phi dp =$

- $P_\phi = \int_{-\infty}^{\infty} \phi^* P \phi dp =$

En (1):

$$\psi(x, t) = \frac{1}{\sqrt{2\pi\hbar}} \int dp \tilde{\psi}(p, t) e^{\frac{i}{\hbar}(px - Et)}$$

$$\left(\frac{e^{\frac{i}{\hbar}(p'x - Et)}}{\sqrt{2\pi\hbar}}, \frac{e^{\frac{i}{\hbar}(px - Et)}}{\sqrt{2\pi\hbar}} \right) = \delta(p' - p)$$

$\uparrow$
 Delta de
 dirac

Nota: $\int \delta(\bar{r} - \bar{a}) f(\bar{r}) d^3\bar{r} = f(\bar{a})$

$$\tilde{\psi}(p, t) = \frac{1}{\sqrt{2\pi\hbar}} \int dx \psi(x, t) e^{-\frac{i}{\hbar}(px - Et_0)}$$

En 3D:

$$\psi_{(t, \bar{x})} = \frac{1}{(\sqrt{2\pi\hbar})^3} \int d^3\bar{P} \psi(\bar{P}, t) e^{\frac{i}{\hbar}(\bar{P} \cdot \bar{x} - Et)}$$

$$\left\{ \frac{e^{\frac{i}{\hbar}(\bar{P} \cdot \bar{x} - Et)}}{(\sqrt{2\pi\hbar})^3} \right\} \cdot \begin{array}{l} \text{Base ortonormal} \\ \text{de ondas planas} \end{array}$$

$$\left(\frac{e^{\frac{i}{\hbar} \bar{P}' \cdot \bar{x}}}{(\sqrt{2\pi\hbar})^3}, \frac{e^{\frac{i}{\hbar} \bar{P} \cdot \bar{x}}}{(\sqrt{2\pi\hbar})^3} \right) = \delta(\bar{P}' - \bar{P}) \begin{cases} \infty; & \bar{P}' = \bar{P} \\ 0; & \bar{P}' \neq \bar{P} \end{cases}$$

$$\tilde{\psi}(\bar{P}, t) = \frac{1}{(\sqrt{2\pi\hbar})^3} \int d^3\bar{x} \, \psi(\bar{x}, t) e^{-\frac{i}{\hbar}(\bar{P} \cdot \bar{x} - Et)}$$

Tarea: Escribir la ecuación de Schrödinger en el espacio momentum.
(Revisar Cohen)

$$\psi \begin{cases} \psi(\bar{x}, t) \\ \phi(\bar{P}, t) \end{cases}$$

$$H\psi = \frac{\hbar}{i} \frac{\partial \psi}{\partial t}$$

$$R\psi \begin{cases} \bar{r} \psi \\ -\frac{\hbar}{i} \nabla_{\bar{P}} \tilde{\psi} \end{cases}$$

El principio de Incertidumbre de Heisenberg

A, B auto adjuntos

$$\sigma_{\psi} A \sigma_{\psi} B \geq \frac{1}{2} |[A, B]_{\psi}|$$

$$[A, B] = 0 \rightarrow [A, B] \psi = 0$$

$$A \underbrace{B \psi} = B A \psi$$

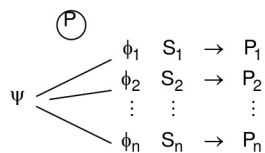
$$\text{Ejm: } [P, R] = \frac{\hbar}{i} \quad \boxed{\sigma_{\psi} P \sigma_{\psi} R \geq \frac{\hbar}{2}}$$

(no conmutan)

¿Qué significa?

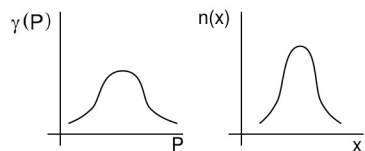
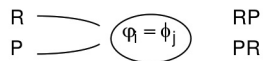
$$R\psi \longrightarrow (\psi, R\psi) \equiv \mathcal{R}_{\psi}$$

$$\begin{array}{c} \textcircled{R} \\ \psi \begin{cases} \varphi_1 & r_1 & \rightarrow & P_1 \\ \varphi_2 & r_2 & \rightarrow & P_2 \\ \vdots & \vdots & & \vdots \\ \varphi_n & r_n & \rightarrow & P_n \end{cases} \end{array}$$



$$(PR - RP)\psi = PR\psi - RP\psi \neq 0$$

$PR\psi \neq RP\psi$ (No hay una base común) (No pueden medirse simultáneamente)



Si en una medida se hace precisa, la otra se hace imprecisa.

Debemos demostrar que:

$$\forall \alpha \in \mathbb{C}, \left\| \frac{\mu}{\|\mu\|} - \alpha \frac{v}{\|v\|} \right\|^2 = 1 + |\alpha|^2 - \frac{\alpha(\mu, v) + \alpha^*(\mu, v)^*}{\|\mu\| \|v\|}$$

$\uparrow$
 $(\in \mathbb{C})$

$$\begin{aligned}
 &\rightarrow \left(\frac{\mu}{\|\mu\|} - \alpha \frac{v}{\|v\|}, \frac{\mu}{\|\mu\|} - \alpha \frac{v}{\|v\|} \right) \\
 &= \frac{(\mu, \mu)}{\|\mu\|^2} - \frac{\alpha(\mu, v)}{\|\mu\| \|v\|} - \frac{\alpha^*(v, \mu)}{\|\mu\| \|v\|} + |\alpha|^2 \frac{(v, v)}{\|v\|^2} \\
 &= 1 - \frac{\alpha(\mu, v) + \alpha^*(\mu, v)^*}{\|v\| \|\mu\|} + |\alpha|^2
 \end{aligned}$$

$$\begin{aligned}
& \left| \alpha - \frac{(\mu, \nu)^*}{\|\mu\| \|\nu\|} \right|^2 = |\alpha|^2 + \frac{|(\mu, \nu)|^2}{\|\mu\|^2 \|\nu\|^2} - \left[\frac{\alpha (\mu, \nu)}{\|\mu\| \|\nu\|} + \frac{\alpha^* (\mu, \nu)^*}{\|\mu\| \|\nu\|} \right] \\
& \rightarrow \left(\alpha - \frac{(\mu, \nu)^*}{\|\mu\| \|\nu\|} \right) \left(\alpha^* - \frac{(\mu, \nu)}{\|\mu\| \|\nu\|} \right) = |\alpha|^2 - \frac{\alpha (\mu, \nu)}{\|\mu\| \|\nu\|} - \frac{\alpha^* (\mu, \nu)^*}{\|\mu\| \|\nu\|} + \frac{|(\mu, \nu)|^2}{(\|\mu\| \|\nu\|)^2} \\
& \rightarrow \left\| \frac{\mu}{\|\mu\|} - \alpha \frac{\nu}{\|\nu\|} \right\|^2 = 1 + \left| \alpha - \frac{(\mu, \nu)^*}{\|\mu\| \|\nu\|} \right|^2 - \frac{|(\mu, \nu)|^2}{\|\mu\|^2 \|\nu\|^2}
\end{aligned}$$

Si tomamos: $\alpha = \frac{(\mu, \nu)^*}{\|\mu\| \|\nu\|}$

$$\Rightarrow |(\mu, \nu)|^2 \leq \|\mu\|^2 \|\nu\|^2$$

$$\Rightarrow \boxed{|(\mu, \nu)| \leq \|\mu\| \|\nu\|} \text{ Teorema de Schwartz}$$

- A, B autoadjuntos:

$$\begin{aligned}
a &= A\psi \\
b &= B\psi
\end{aligned}
\begin{matrix} \searrow \\ \nearrow \end{matrix} \in \mathbb{R}$$

$$\begin{aligned}
a &= A\psi \\
b &= B\psi
\end{aligned}$$

$$a = (\psi, A\psi)$$

$$a^* = (\psi, A\psi)^* = (A\psi, \psi)$$

(A pasa con su adjunta, pero como A es autoadjunta $\rightarrow \text{Adj } A = A$)

$$= (\psi, A\psi) = a$$

$$\text{Como } a^* = a \Rightarrow a \in \mathbb{R}$$

- $\|(A-a)\psi\| \|(B-b)\psi\| \geq \left| \left((A-a)(B-b)\psi, \psi \right) \right| = \left| \left(\psi, (A-a)(B-b)\psi \right) \right|$
 $\rightarrow \left| \left((A-a)\psi, (B-b)\psi \right) \right| \leq \|(A-a)\psi\| \|(B-b)\psi\|$
 $\left| \left((A-a)(B-b)\psi, \psi \right) \right|$

- $[P, Q] = PQ - QP$ conmutador

$$\{P, Q\} = PQ + QP \text{ anticonmutador}$$

$$\text{Verifique } \{P, Q\}_\psi \in \mathbb{R}$$

Para operadores autoadjuntos:

$[P, Q]_\psi$ es imaginario puro

$$\begin{aligned} [P, Q]_\psi &= (\psi, (PQ - QP) \psi) \\ &= (\psi, PQ\psi) - (\psi, QP\psi) \end{aligned}$$

$$\begin{aligned} [P, Q]_\psi^* &= (PQ\psi, \psi) - (QP\psi, \psi) \\ &= (\psi, QP\psi) - (\psi, PQ\psi) \\ &= -(\psi, [P, Q]\psi) \end{aligned}$$

$\Rightarrow [P, Q]_\psi$ es imaginario puro

Verifique:

$$|[A, B]_\psi + \{A - a, B - b\}_\psi|^2 = \underbrace{|[A, B]_\psi|^2}_{\text{Imag (ib)}} + \underbrace{|\{A - a, B - b\}_\psi|^2}_{\text{real (a)}} \dots (1)$$

Verifique:

$$2((A - a)(B - b))_\psi = [A, B]_\psi + \{A - a, B - b\}_\psi \dots (2)$$

(2) en (1):

$$4((A - a)(B - b))_\psi^2 = |[A, B]_\psi|^2 + |\{A - a, B - b\}_\psi|^2$$

$$4\|(A - a)\psi\|^2 \|(B - b)\psi\|^2 \geq |[A, B]_\psi|^2 + |\{A - a, B - b\}_\psi|^2 \geq |[A, B]_\psi|^2$$

$$2\|(A - a)\psi\| \|(B - b)\psi\| \geq |[A, B]_\psi|$$

$$\|(A - a)\psi\|^2 = ((A - a)\psi, (A - a)\psi) = (\psi, (A - a)^2 \psi) = (\sigma_\psi A)^2$$

$$\|(B - b)\psi\|^2 = (\sigma_\psi B)^2$$

$$\sigma_\psi A = \|(A - a)\psi\|, \sigma_\psi B = \|(B - b)\psi\|$$

$$\Rightarrow \sigma_{\psi} A \sigma_{\psi} B \geq \frac{1}{2} |[A, B]_{\psi}|$$

$$\Delta_{\psi} R \Delta_{\psi} P \geq \frac{\hbar}{2}$$

Verificar:

- $[B, A] = -[A, B]$
- $[A + B, C + D] = [A, C] + [A, D] + [B, C] + [B, D]$
- $[\alpha A, \beta B] = \alpha \beta [A, B]$
- $[AB, C] = A[B, C] + [A, C]B$
- $[A, BC] = [A, B]C + B[A, C]$
- $[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$

$$\begin{aligned} \overleftarrow{[AB, C]} &= A[B, C] + [A, C]B \\ \Rightarrow [AB, C] &= ABC - CAB \\ &= ABC - ACB + ACB - CAB \\ &= A[B, C] + [A, C]B \end{aligned}$$

Sea el hamiltoniano del oscilador armónico (unidimensional).

Verificar que la energía E , de la ecuación estacionaria $H\psi = E\psi$, queda expresada por

$$E = \frac{1}{2m} \|P\psi\|^2 + m \frac{\omega^2}{2} \|R\psi\|^2$$

$$\begin{aligned} H &= \frac{P^2}{2m} + m \frac{\omega^2}{2} R^2 \\ E &= (\psi, H\psi) = \left(\psi, \frac{P^2}{2m} \psi \right) + \left(\psi, m \frac{\omega^2}{2} R^2 \psi \right) \\ &= \frac{1}{2m} (P\psi, P\psi) + m \frac{\omega^2}{2} (R\psi, R\psi) \\ &= \frac{1}{2m} \|P\psi\|^2 + m \frac{\omega^2}{2} \|R\psi\|^2 \end{aligned}$$

- $\|A\psi\|^2 = A\psi^2 + (\sigma_\psi A)^2$

$$E = \frac{1}{2m} P_\psi^2 + m \frac{\omega^2}{2} R_\psi^2 + \frac{1}{2m} (\sigma_\psi P)^2 + \\ + m \frac{\omega^2}{2} (\sigma_\psi R)^2$$

- Para $x, z \in \mathbb{R} \rightarrow x^2 + z^2 \geq 2xz$

$$E \geq \frac{1}{2m} P_\psi^2 + m \frac{\omega^2}{2} R_\psi^2 + 2 \frac{1}{\sqrt{2m}} \frac{\sqrt{m\omega^2}}{\sqrt{2}}$$

$$\sigma_\psi P \sigma_\psi R$$

$$E \geq \frac{P_\psi^2}{2m} + m \frac{\omega^2}{2} R_\psi^2 + \underbrace{\omega \sigma_\psi P \sigma_\psi R}_{\hbar/2} \geq \frac{\hbar \omega}{2}$$

$$\left(E \geq \frac{\hbar \omega}{2} \right) \text{ min de energía}$$

Teorema de Ehrenfest

$$\frac{d}{dt} A_\psi = -\frac{i}{\hbar} [A, H]_\psi + \left(\frac{\partial A}{\partial t}\right)_\psi$$

Sea A un operador lineal, la derivada temporal de su valor esperado es:

$$\frac{d}{dt} (\psi, A\psi) = \frac{d}{dt} \int \psi^* A\psi d^3\vec{r} = \left(\frac{\partial \psi}{\partial t}, A\psi\right) + (\psi, A\frac{\partial \psi}{\partial t}) + (\psi, \frac{\partial A}{\partial t}\psi)$$

$$i\hbar \frac{\partial \psi}{\partial t} = H\psi \rightarrow \frac{\partial \psi}{\partial t} = -\frac{i}{\hbar} H\psi$$

$$\begin{aligned} \frac{d}{dt} (\psi, A\psi) - (\psi, \frac{\partial A}{\partial t}\psi) &= \left(\frac{\partial \psi}{\partial t}, A\psi\right) + (\psi, A\frac{\partial \psi}{\partial t}) \\ &= \frac{i}{\hbar} (H\psi, A\psi) - \frac{i}{\hbar} (\psi, AH\psi) \\ &= -\frac{i}{\hbar} (\psi, [A, H]\psi) \end{aligned}$$

$$\boxed{\frac{d}{dt} A_\psi = -\frac{i}{\hbar} [A, H]_\psi + \left(\frac{\partial A}{\partial t}\right)_\psi}$$

Ejm

$$1) [x, H] = [x, \frac{p^2}{2m} + V] = [x, \frac{p^2}{2m}] = i\frac{p\hbar}{m}$$

$$\boxed{[x, p] = i\hbar}$$

$$\frac{d}{dt} x_\psi = \frac{p_\psi}{m}$$

$$\boxed{p = \hbar \frac{\partial}{\partial x}}$$

$$2) [p, H] = [p, \frac{p^2}{2m} + V] = [p, V] = -i\hbar \nabla V$$

$$\boxed{[x, p^2] = p[x, p] + [x, p]p}$$

$$\begin{aligned} [p, V]\psi &= p(V\psi) - Vp\psi \\ &= \underline{pV\psi} + \underline{Vp\psi} - \underline{Vp\psi} \end{aligned}$$

$$\frac{d}{dt} P_\psi = - \nabla V = F$$

Si el operador H no depende explícitamente del tiempo,
 como $H = \frac{p^2}{2m} + V(R)$, entonces:

$$\frac{d}{dt} H_\psi = - \frac{i}{\hbar} [H, H]_\psi = 0 \rightarrow H_\psi = \text{cte}$$

$\frac{\partial H}{\partial t} = 0$

Ejm $P_\psi = m \dot{X}_\psi \rightarrow \dot{P}_\psi = m \ddot{X}_\psi$

$$\dot{P}_\psi = - \left(\frac{\partial V}{\partial x} \right)_\psi$$

$$m \ddot{X}_\psi = - \left(\frac{\partial V}{\partial x} \right)_\psi = -K X_\psi \rightarrow \ddot{X}_\psi = - \frac{K}{m} X_\psi$$

$$X_{\psi} = A \sin \omega t + B \cos \omega t$$

$$\dot{X}_{\psi} = A\omega \cos \omega t - B\omega \sin \omega t$$

$$\ddot{X}_{\psi} = -A\omega^2 \sin \omega t - B\omega^2 \cos \omega t$$

$$= -\omega^2 X_{\psi}, \quad \omega = \sqrt{\frac{k}{m}}$$

NOTA

$$[X, P] = i\hbar \rightarrow XP\psi - P(X\psi) = i\hbar \psi$$

$$\underline{XP\psi} - (PX)\psi - \underline{XP\psi} = i\hbar \psi$$

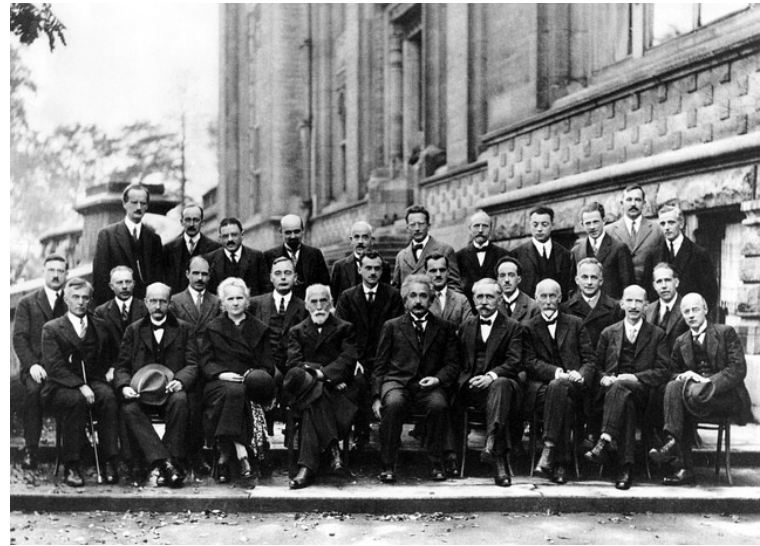
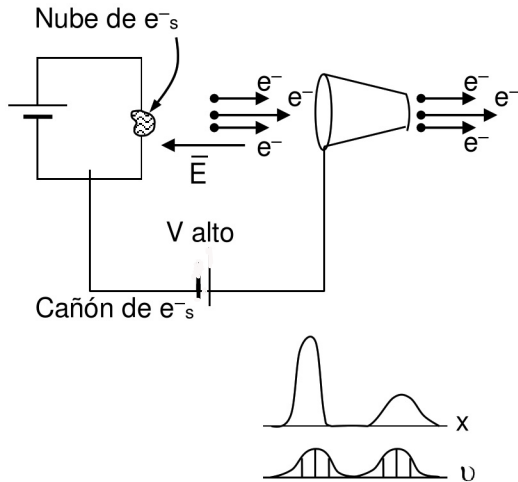
$$-(PX)\psi = i\hbar \psi \quad P \equiv \frac{\hbar}{i} \frac{\partial}{\partial x}$$

$$[X, P]\tilde{\psi} = i\hbar \tilde{\psi} \rightarrow X(P\tilde{\psi}) - P(X\tilde{\psi}) = i\hbar \tilde{\psi}$$

$$(XP)\tilde{\psi} + \underline{PX\tilde{\psi}} - \underline{PX\tilde{\psi}} = i\hbar \tilde{\psi}$$

$$X \equiv -\frac{\hbar}{i} \frac{\partial}{\partial P}$$

Partícula Libre



Partícula Libre:

- 1) Distribución gaussiana de posición y momentum en t .
- 2) Avance con $V = \text{cte}$ de R_ψ es acompañado del ensanchamiento de la curva de distribución de posición.
- 3) Inalterabilidad de la curva de distribución de momentums.

Hipótesis de Equivalencia:

Haz de N electrones	$\leftrightarrow$	e^- haciendo el recorrido N veces
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$$-\frac{\hbar^2}{2m} \nabla^2 \psi + \frac{\hbar}{i} \partial \psi = 0$$

$$n_t(\vec{r}) = e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

$$\psi_t(\vec{r}) = c \int d^3\vec{x} \phi(\vec{k}) e^{i(\vec{k} \cdot \vec{r} - \omega t)}$$

Coef. Fourier $\omega = \frac{\hbar k^2}{2m}$

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int d^3\vec{r} \psi(0, \vec{r}) e^{-i\vec{k} \cdot \vec{r}}$$

En 1D:

$$\psi(x, t) = \frac{1}{\sqrt{2\pi}} \int dk \phi(k) e^{i(xk - \omega t)}$$

$$\hbar k$$

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int dx \psi(\overset{\curvearrowright}{0}, x) e^{-ikx}$$

Recuerde que:

$$v(x) = e^{-ax^2}$$

$$\phi(k) = \int_{-\infty}^{\infty} dx e^{-ax^2} e^{-ikx} \left(\frac{1}{\sqrt{2\pi}} \right)$$

$$-ax^2 - ikx = -a \left(x^2 + \frac{ik}{a} x \right)$$

$$= -a \left[\left(x + \frac{ik}{2a} \right)^2 + \frac{k^2}{4a^2} \right]$$

$$\phi(k) = e^{-\frac{k^2}{4a}} \underbrace{\int dx e^{-a \left(x + \frac{ik}{2a} \right)^2}}_{\substack{\in \mathbb{C} \\ \sqrt{\frac{\pi}{a}}}}$$

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \sqrt{\frac{\pi}{a}} e^{-\frac{k^2}{4a}}$$

$$P_v(x) = \frac{\hbar}{i} \frac{d}{dx} e^{-ax^2} = \frac{\hbar}{i} (-2ax) e^{-ax^2}$$

$$P_\psi = \int_{-\infty}^{\infty} v^* P_v dx = -\frac{2a\hbar}{i} \int_{-\infty}^{\infty} x e^{-2ax^2} dx = 0$$

Esto indica que $P_v = 0$, lo que quiere decir que  (no representa a una partícula en movimiento)

Se encuentra en reposo

$$\begin{aligned} n(t, \vec{r}) &= e^{i(\vec{k} \cdot \vec{r} - \omega_k t)} \\ v(x) &= e^{-ax^2} \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} \begin{array}{l} \text{no son} \\ \text{compatibles} \end{array}$$

P_ψ, R_ψ son reales

$$\eta(x) = \underline{A(x)} e^{ikx} \rightarrow \frac{d\eta}{dx} = A' e^{ikx} + ikA e^{ikx}$$

$$P_\psi = \frac{\hbar}{i} \int \eta^* \frac{d\eta}{dx} dx = \frac{\hbar}{i} \int_{-\infty}^{\infty} A^* A' dx + \frac{\hbar}{i} ik \int_{-\infty}^{\infty} A^* A dx$$

Se debe elegir A de modo que:

$$\int dx A^* A' = 0$$

$$A = e^{-ax^2}$$

Tomaremos

$$\psi(x) = B e^{-ax^2 + ibx}$$

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int dx \psi(x) e^{-ikx}$$

$$\phi(k) = \frac{B}{\sqrt{2\pi}} \int dx e^{-ax^2 + i(b-k)x}$$

$$-ax^2 + i(b-k)x = -a \left[\left(x - \frac{i(b-k)}{2a} \right)^2 + \frac{(b-k)^2}{4a^2} \right]$$

$$\phi(k) = B \frac{e^{-\frac{(b-k)^2}{4a}}}{\sqrt{2\pi}} \int dx e^{-a \left(x - \frac{i(b-k)}{2a} \right)^2}$$

$$\rightarrow \phi(k) = \frac{B}{\sqrt{2a}} e^{-\frac{(b-k)^2}{4a}}$$

Paquete de onda $\equiv$ partícula libre

$$\psi(t, x) = \frac{1}{\sqrt{2\pi}} \int dk \phi(k) e^{i(kx - \omega_k t)}; \quad \omega_k = \frac{\hbar k^2}{2m}$$

Argumento exponente

$$-\frac{(k-b)^2}{4a} + i\left(kx - \frac{\hbar k^2}{2m}t\right)$$

$$= -\frac{N}{4a}\left[k - \frac{b+2ia}{N}x\right]^2 - \frac{a}{N}\left(x - \frac{ib}{2a}\right)^2 - \frac{b^2}{4a^2}$$

$$N = 1 + i\alpha$$

$$\alpha = 2\frac{\hbar a}{m}t$$

$$\psi(t, x) = \frac{B}{N^{1/2}} e^{-\frac{a}{N}\left(x - \frac{ib}{2a}\right)^2 - \frac{b^2}{4a^2}}$$

$$-\frac{a}{N}\left(x - \frac{ib}{2a}\right)^2 - \frac{b^2}{4a^2}$$

$$= \left(-\frac{a}{|N|^2}\right) \left[\left(x - \frac{\alpha b}{2a}\right)^2 + i\left(\frac{a}{|N|^2}\right) \cdot \left[\alpha\left(x^2 - \frac{b^2}{4a^2}\right) + \frac{b}{a}x \right] \right]$$

$$\rho(t, x) = |\psi(t, x)|^2$$

$$= \frac{B^2}{|N|} e^{-\frac{2a}{|N|^2}\left(x - \frac{\hbar b t}{m}\right)^2}$$

$$R_\psi = (\psi, R\psi) = \frac{\hbar b t}{m} \quad \begin{array}{l} \text{(por 1er teorema de} \\ \text{Ehrenfest,} \\ \text{encuentras } P_\psi) \end{array}$$

$$\rightarrow P_\psi = (\psi, P\psi) = \hbar b$$

$$\sigma_\psi R = \frac{|N|}{2\sqrt{a}} \quad N = 1 + i\alpha t$$

$$\sigma_\psi P = \sqrt{a} \hbar \quad \xi = \frac{2\hbar a}{m}$$

$$\sigma_\psi R \sigma_\psi P = \frac{|N| \hbar}{2}$$

(cumple principio de incertidumbre Heisenberg)

$$\frac{1}{\sqrt{2\pi}} \frac{B}{\sqrt{2a}} \sqrt{\frac{\pi}{N}} \sqrt{4a} = \frac{B}{N^{1/2}}$$